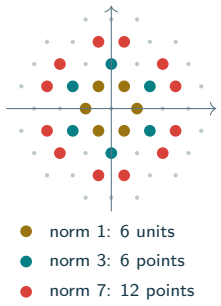
NUMBER THEORY LADDER 8 OF 9

Primes mod 3 and the Eisenstein integers

A prime is a value of $a^2 - ab + b^2$ exactly when it is 1 mod 3, which is exactly when it splits in the Eisenstein integers.

1 / THE RULE AND THE PICTURE

In $\mathbb{Z}[\omega] = \{a + b\omega\}$ the norm is $a^2 - ab + b^2$, and points of equal norm form hexagonal rings



With $\omega = e^{2\pi i/3}$ we have $N(a + b\omega) = (a + b\omega)(a + b\bar{\omega}) = a^2 - ab + b^2$, and the number of lattice points of norm n is

$$\boxed{r(n) = 6\left(d_1(n) - d_2(n)\right)}$$

where d_1, d_2 count divisors that are 1 and 2 modulo 3. For 7 the divisors are $1, 7$, both $1 \bmod 3$, giving 12 points, and for 3 we get 6.

2 / SPLIT, INERT AND RAMIFIED

Every prime of $\mathbb{Z}$ again behaves in one of three ways

type	condition	examples	norm form
split	$p \equiv 1 \pmod 3$	$7 = (3+\omega)(3+\bar{\omega})$, $13 = (4+\omega)(4+\bar{\omega})$	$p = a^2 - ab + b^2$
inert	$p \equiv 2 \pmod 3$	$2, 5, 11, 17$ stay prime	no representation
ramified	$p = 3$	$3 = -\omega^2(1-\omega)^2$	$3 = 2^2 - 2 + 1$

The primes $1 \bmod 3$ up to 100 are $7 = 3^2 - 3 + 1$, $13 = 4^2 - 4 + 1$, $19 = 5^2 - 10 + 4$, $31 = 6^2 - 6 + 1$, $37 = 7^2 - 21 + 9$, $43 = 7^2 - 7 + 1$, $61 = 9^2 - 36 + 16$, $67 = 9^2 - 18 + 4$, $73 = 9^2 - 9 + 1$, $79 = 10^2 - 30 + 9$, $97 = 11^2 - 33 + 9$.

3 / WHY $1 \bmod 3$ / THE EQUATION $x^2 + x + 1 \equiv 0$

A prime splits exactly when $x^2 + x + 1$ has a root modulo p

p	roots of $x^2 + x + 1$
7	2, 4
13	3, 9
5	none
11	none

A root is a cube root of unity modulo p, which exists exactly when $3 \mid p - 1$. Then $p \mid x^2 + x + 1 = (x - \omega)(x - \bar{\omega})$ while p divides neither factor, so p is not prime in $\mathbb{Z}[\omega]$ and splits. For $p = 7$ the root 2 gives $2^2 + 2 + 1 = 7$.

4 / WHICH NUMBERS ARE NORMS

Primes $2 \bmod 3$ must appear to an even power, and norms multiply

The values of $a^2 - ab + b^2$ are $1, 3, 4, 7, 9, 12, 13, 16, 19, 21, \ldots$ while $2, 5, 6, 8, 10, 11$ never occur. Norms multiply, so $7 \cdot 13 = 91$ with $(3+\omega)(4+\omega) = 11 + 6\omega$ gives $91 = 11^2 - 11 \cdot 6 + 6^2$. The same form appears in the cosine rule for a $60°$ angle, $c^2 = a^2 - ab + b^2$, so a triangle with sides $5, 8, 7$ has a $60°$ angle opposite 7.

5 / THE RACE BETWEEN THE CLASSES

Primes $2 \bmod 3$ lead for a very long way

	$1 \bmod 3$	$2 \bmod 3$	
up to 100	11	13	$2 \bmod 3$ ahead
up to 1000	80	87	$2 \bmod 3$ ahead
up to 10 000	611	617	$2 \bmod 3$ ahead

The prime 2 is counted with the inert class and 3 is excluded. The $1 \bmod 3$ count first overtakes at 608 981 813 029, far beyond the mod 4 case at 26 861.

ASIDE / THE HEXAGON

The six units $\pm 1, \pm\omega, \pm\omega^2$ are the vertices of the hexagon, so each prime has six associates.

LINKS TO OTHER SHEETS

The square lattice version is *Primes mod 4 and the Gaussian integers*; the cube root of unity appears in *Cyclotomic fields and the arcs of inscribed polygons*; the last sheet is *Fibonacci modulo p and the golden ratio*.

Convention. $N(a + b\omega) = a^2 - ab + b^2$; counts exclude 3; $\mathbb{Z}[\omega]$ has unique factorisation up to the six units.